SMART DUSTBIN

S.E. Capstone Project Report submitted in partial fulfilment of the requirements for the degree of

Bachelor of Technology (B. Tech.)

In

COMPUTER ENGINEERING

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CERTIFICATE

This is to certify that the S.E. project entitled “Smart Dustbin” is a bonafide work of “simran ament”, “arya sankhe”, “mahek bhanushali” (EU2252002,EU2252010,EU2252011) submitted to University of Mumbai in partial fulfilment of the requirement for the award of the degree of “Bachelor of Technology” in “Computer Engineering” during the academic year 2026 – 2027.

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S.E. CAPSTONE PROJECT REPORT APPROVAL

This project report entitled **Smart Dustbin** by **Simran Ament, Arya Sankha, Mahek Bhanushali** is approved for the degree of **Bachelor of Technology** in **Computer Engineering** from University of Mumbai.

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ABSTRACT

The **Smart Dustbin** is a basic automated waste-management system designed to make waste disposal more convenient, hygienic, and user-friendly. The main purpose of this project is to reduce direct contact with the dustbin while throwing waste. In this project, an **ultrasonic sensor** is used to detect the presence of a person or object near the dustbin. When an object comes within a particular distance, the sensor sends a signal to the system, which activates a **servo motor** to open the lid automatically. After a short period, the lid can return to its original position.

The project uses a **breadboard, ultrasonic sensor, jumper wires, servo motor, and LED lights** as its main components. The breadboard provides a convenient platform for connecting the components, while jumper wires are used to establish electrical connections. The ultrasonic sensor helps in detecting objects without physical contact, and the servo motor controls the movement of the dustbin lid. LED lights are used to indicate the working or operating status of the system.

This project demonstrates how simple electronic components can be combined to create an automated and useful everyday device. The Smart Dustbin can help maintain cleanliness by encouraging people to dispose of waste properly and by reducing unnecessary touching of the dustbin. Although this is a basic prototype, it provides a foundation for developing more advanced smart waste-management systems in the future.

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
cm	Centimetre
ESP8266	ESP8266 wifi microcontroller
GPIO	General Purpose Input/Output
HC-SR04	Ultrasonic distance sensor module
IDE	Integrated Development Environment
IoT	Internet of Things
LED	Light Emitting Diode
SDG	Sustainable Development Goal
USB	Universal Serial Bus
Wi-Fi	Wireless Fidelity

CHAPTER 1
INTRODUCTION

The Smart Dustbin is an automated dustbin designed to make waste disposal easy and hygienic. It uses an ultrasonic sensor to detect objects and a servo motor to open the lid automatically .This project reduces direct contact with the dustbin and helps maintain cleanliness

1.1 Background of the Study

Waste management is an important issue because improper disposal of garbage can cause bad smell, pollution, germs, and unhealthy surroundings. Traditional dustbins usually require people to open the lid manually and also need regular checking to prevent overflow. This creates a need for a more hygienic and convenient waste-disposal system.

A smart dustbin is a basic waste bin improved with electronic components to provide automatic operation. In many designs, an ultrasonic sensor detects the presence of a person's hand or an object near the bin. The signal is processed by a controller such as Arduino, which then operates a servo motor to open and close the lid automatically. This reduces the need to touch the dustbin.

Smart dustbins can also be designed to monitor the amount of waste inside the bin. More advanced systems may use LEDs, displays, buzzers, or wireless communication to indicate when the bin is full. However, a basic smart dustbin mainly focuses on automatic lid operation, hygiene, and ease of use.

For a basic project, components such as a breadboard, ultrasonic sensor, jumper wires, servo motor, and LED lights can be used to demonstrate the working of a smart dustbin. The ultrasonic sensor detects an object, the control circuit processes the signal, and the servo motor moves the lid. LED lights can provide a simple visual indication of the system's operation.

Overall, the background study shows that smart dustbins combine waste management with simple automation. They can reduce human contact with waste, improve cleanliness, and make waste disposal more convenient while remaining relatively simple and low-cost to build.

1.2 Motivation

The main motivation behind the Smart Dustbin project is to create a cleaner, more hygienic, and convenient way of disposing of waste. Traditional dustbins

require users to touch the lid, which can transfer germs and bacteria. They may also be left open, causing bad smell and attracting insects.

The Smart Dustbin uses automation to reduce these problems. An ultrasonic sensor detects when a person brings their hand or an object near the dustbin. The servo motor then automatically opens the lid, allowing waste to be thrown without touching the bin. LED lights can be used to indicate the working of the system.

The project is also motivated by the increasing need for proper waste management and smart technology in homes, schools, offices, and public places. It demonstrates how simple and affordable electronic components can be used to solve everyday problems.

1.3 Problem Statement

Traditional dustbins require people to open and close the lid manually while disposing of waste. This can lead to direct contact with the dustbin, making the process unhygienic and increasing the chances of spreading germs. It can also be inconvenient, especially when the user's hands are occupied or dirty. The proposed Smart Dustbin provides an automatic and touchless solution for waste disposal. The system uses an ESP8266, ultrasonic sensor, and servo motor. When the ultrasonic sensor detects an object near the dustbin, the

ESP8266 processes the signal and operates the servo motor to open the lid automatically. After a short period, the lid closes automatically. The main aim

of this project is to make waste disposal more hygienic, convenient, and user-friendly while reducing unnecessary physical contact with the dustbin.

1.4 Objectives

- To develop an automatic dustbin:
- To design a basic smart dustbin that can open and close its lid automatically without requiring manual operation.
- To detect objects near the dustbin:
- To use an ultrasonic sensor to detect when a person brings their hand or waste near the dustbin.
- To control the system using ESP8266:
- To use the ESP8266 microcontroller to receive the sensor input and control the operation of the dustbin.
- To automate the dustbin lid:
- To use a servo motor to open the lid when an object is detected and close it after a short time.
- To reduce physical contact:
- To minimize the need to touch the dustbin while disposing of waste, making the process more hygienic.
- To improve convenience:
- To make waste disposal easier and more user-friendly through automatic operation.
- To create a simple and low-cost system:
- To develop an easy-to-build smart dustbin using commonly available electronic components.
- To demonstrate automation:
- To demonstrate how sensors, a microcontroller, and a motor can work together to automate a simple everyday task.

1.5 Scope and Limitations

Scope:- The Smart Dustbin is an innovative project designed to improve cleanliness and hygiene using modern technology. It uses an ESP8266 microcontroller, an ultrasonic sensor, and a servo motor to operate automatically.

1. Automatic Lid Operation: The dustbin lid opens automatically when a hand or object comes near the sensor and closes after a certain time.

2. Improved Hygiene: It reduces physical contact with the dustbin, helping to prevent the spread of germs and maintain cleanliness.

3. Use in Different Places: The Smart Dustbin can be used in homes, colleges, schools, offices, hospitals, shopping malls, and public places.

4. Time-Saving: The automatic mechanism makes waste disposal easier and more convenient for users.

5. Environmental Cleanliness: It encourages proper waste disposal and helps maintain a clean and hygienic environment.

Limitations. Although the Smart Dustbin is useful, it has some limitations that should be considered.

1. Power Dependency: The system requires a proper power supply to operate the sensor, microcontroller, and servo motor.

2. Sensor Accuracy: The ultrasonic sensor may sometimes give incorrect readings due to object position, distance, or environmental conditions.

3. Limited Storage Capacity: The dustbin has limited space and must be emptied regularly when it becomes full.

4. Maintenance Required: The sensor, servo motor, wires, and other electronic components require occasional inspection and maintenance.

CHAPTER 2

REVIEW OF LITERATURE

2.1 Introduction

A literature review provides an overview of the existing work and technologies related to a particular field and helps identify areas where improvements can be made. For this project, the review focuses on existing automatic and smart dustbin systems, their working methods, and the use of sensors and microcontrollers for touchless waste disposal. The project mainly focuses on automatic lid operation using an ultrasonic sensor, ESP8266, and servo motor. Basic concepts related to ultrasonic sensing, ESP8266 programming, servo motor control, and automation were studied from standard textbooks, research papers, and other technical references.

2.2 Review of Existing Systems

Traditional dustbins generally require users to manually open and close the lid while disposing of waste. Such systems involve direct contact with the dustbin and may be less hygienic and convenient. Automatic dustbins have been developed to overcome this problem by using sensors to detect the presence of a person or object near the dustbin. When an object is detected, the lid can be opened automatically using a motor.Various automatic dustbin systems use components such as ultrasonic sensors, infrared sensors, Arduino boards, ESP8266, and servo motors. Among these, ultrasonic sensors are commonly used for detecting objects based on distance. Microcontrollers process the sensor output and control the opening and closing mechanism of the dustbin

2.3 Review of Research Papers

Several research projects have proposed sensor-based automatic dustbins for improving hygiene and convenience during waste disposal. These systems generally use a distance sensor to detect the presence of an object near the dustbin and a motor mechanism to operate the lid. Some systems are based on Arduino microcontrollers, while others use ESP8266 or similar development boards. The use of ESP8266 provides a compact and programmable solution for controlling the sensor and servo motor. In the proposed Smart Dustbin, the ultrasonic sensor detects an object placed near the dustbin. The ESP8266

receives this information and operates the servo motor to open the lid automatically. After a short delay, the lid returns to its original position.

2.4 Research Gap

From the study of existing systems, it can be observed that many smart dustbin projects include additional features such as waste-level monitoring, mobile notifications, IoT dashboards, and waste classification. However, these features can increase the complexity and cost of the system.

The proposed project focuses on a simple and low-cost automatic dustbin. It concentrates on the basic requirement of touchless waste disposal. The system uses an ultrasonic sensor for object detection, an ESP8266 as the main controller, and a servo motor for automatic lid movement. This provides a simple, practical, and easy-to-understand solution for automatic waste disposal while reducing direct contact with the dustbin. Waste management is an important part of maintaining a clean and healthy environment. Traditional dustbins are simple to use, but they require manual operation and regular monitoring. People have to touch the lid while disposing of waste, which may increase the possibility of spreading germs and bacteria. Open dustbins can also produce bad smells and attract insects. Because of these problems, researchers have developed smart dustbins that use sensors and automation.

Many existing smart dustbin projects use technologies such as IoT, cloud computing, cameras, mobile applications, GPS, and multiple sensors. These technologies provide advanced features such as monitoring the waste level, sending notifications, identifying different types of waste, and remotely managing the dustbin. However, these systems can be expensive and complicated to develop. They may also require continuous internet connectivity, additional software, and regular maintenance.

CHAPTER 3
SYSTEM ARCHITECTURE

3.1 System Overview

The Smart Dustbin is a basic hardware-based system designed to make the process of disposing of waste more convenient and hygienic. The main purpose of the project is to open the dustbin lid automatically when a person brings their hand or an object near the dustbin. This reduces the need to touch the dustbin manually.The system uses an ESP8266 NodeMCU as the main controller. A sensor is used to detect the presence of a hand or object near the dustbin. When the sensor detects an object within the specified distance, the ESP8266 processes the sensor signal and controls a servo motor. The servo motor rotates the lid to an open position. After a short delay, the lid returns to its closed position automatically.The system consists of three main sections: input, processing, and output. The sensor acts as the input device, the ESP8266 acts as the processing and control unit, and the servo motor acts as the output device.The project is designed as a simple automatic dustbin and does not include advanced features such as waste-level monitoring, mobile notifications, cloud monitoring, waste classification, or automatic waste collection. The main focus is on providing a touch-free method of opening and closing the dustbin lid.The system can be powered using a suitable power supply. The ESP8266 receives the sensor input and generates the required control signal for the servo motor. The

3.2 System Architecture Diagram

The system architecture represents the connection and interaction between the major components of the Smart Dustbin. The basic architecture consists of:

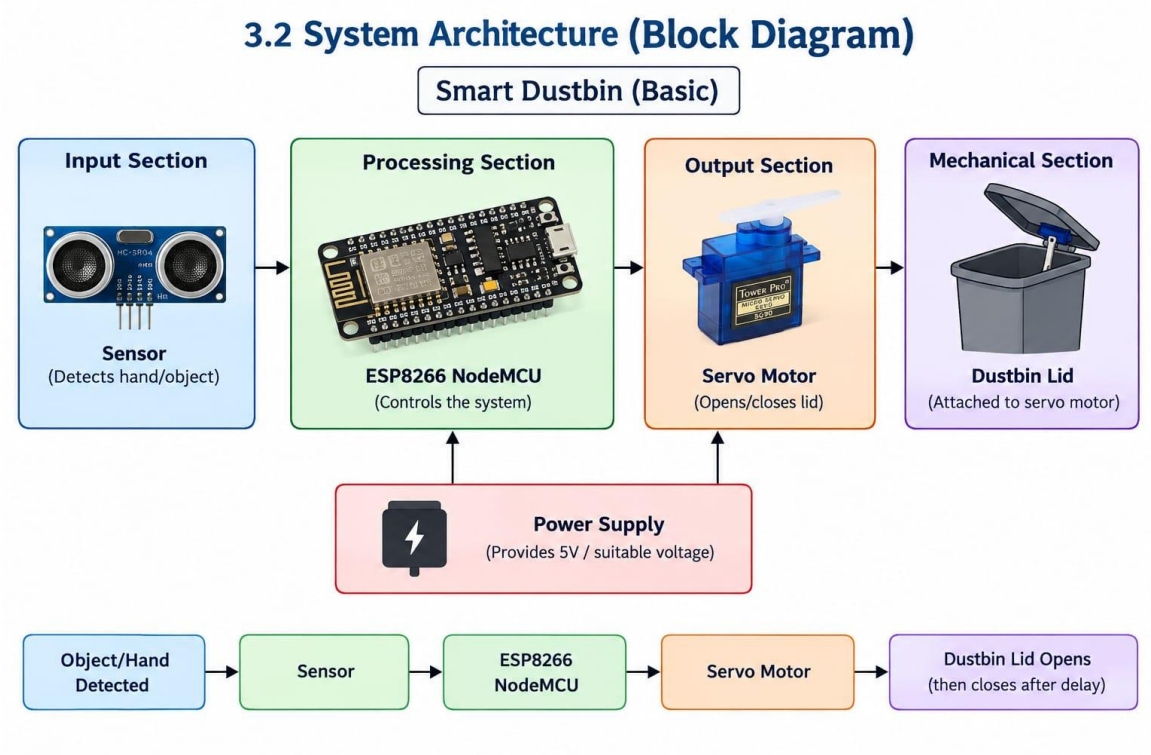


Figure 3.1: System Architecture of Smart Dustbin

3.3 Module Description

Aqua Sentry is divided into the following modules:

- **Sensing Module:** triggers the HC-SR04 and times the returned echo pulse to find the distance to the water surface.
- **Level-Calculation Module:** converts the distance into a water-level percentage using the calibrated empty and full distances of the tank.
- **Display Module:** shows the level %, a level bar and a status line on the 128×64 OLED.
- **Connectivity and Alert Module:** connects the ESP32 to Wi-Fi (or starts the fallback hotspot), serves the dashboard and JSON data over HTTP, and raises the full-tank and low-water alerts when the level crosses the configured thresholds.

CHAPTER 4
REQUIREMENT ANALYSIS

Requirement analysis identifies the hardware, software, and functional requirements needed to develop a Smart Dustbin. The main purpose of the project is to provide a simple and hygienic method of disposing of waste without touching the dustbin lid.

4.1 Hardware Requirements

The following hardware components are required for the development of the basic Smart Dustbin:

Table 4.1: Hardware Requirements

Sr. No.	Hardware Component	Purpose
1	ESP8266 NodeMCU	Main controller of the system
2	Ultrasonic Sensor	Detects the presence of a hand or object
3	Servo Motor	Opens and closes the dustbin lid
4	Dustbin Container	Used for collecting waste
5	Dustbin Lid	Opens and closes automatically
6	Jumper Wires	Used for electrical connections
7	Breadboard/PCB	Used for connecting the components during development
8	Power Supply/USB Cable	Provides power to the system
9	Connecting/Mechanical Arrangement	Connects the servo motor to the lid

4.2 Software Requirements

The Smart Dustbin requires simple software to control the sensor and servo motor. The main software used for the project is **Arduino IDE**. It is used to write, compile, and upload the program to the microcontroller. The program is written using **Embedded C/C++**. It reads the information from the ultrasonic sensor and checks whether an object or hand is present near the dustbin. When the sensor detects an object within a particular distance, the program sends a signal to the **servo motor**. The servo motor then rotates and opens the dustbin lid automatically. After a short delay, the program instructs the servo motor to return to its original position and close the lid. If an LED is

used, the program can also control the LED as an indication of the system's operation.

Sr. No.	Software	Purpose
1	**Arduino IDE**	To write and upload the program
2	**Embedded C/C++**	To program the controller
3	**Servo Library**	To control the servo motor
4	**UltrasonicSensor Code**	To detect a nearby object
5	**Control Program**	To control the opening and closing of the lid
6	**USB Driver**	To connect the controller with a computer

Table 4.2: Software Requirements

4.3 Functional Requirements

The functional requirements describe the **main functions that the basic Smart Dustbin should perform**. The system should work automatically and provide touch-free waste disposal.

1. **Object Detection:** The ultrasonic sensor should detect a hand or object near the dustbin.
2. **Distance Measurement:** The sensor should measure the distance between the object and the dustbin.
3. **Automatic Lid Opening:** When an object is detected within the set distance, the servo motor should open the lid automatically.
4. **Automatic Lid Closing:** After a short delay, the servo motor should return the lid to its original position.
5. **Touch-Free Operation:** The user should be able to dispose of waste without touching the dustbin lid.
6. **LED Indication:** The LED should provide a visual indication when the system is working, if included.
7. **Continuous Operation:** The system should continuously monitor the sensor while power is supplied.
8. **Quick Response:** The lid should open quickly when an object is detected.
9. **Simple Control:** The microcontroller should process the sensor signal and control the servo motor automatically.
10. **Power Operation:** All components should operate properly when connected to the required power supply.

4.4 Non-Functional Requirements

Non-functional requirements describe the **quality, performance, and usability** of the basic Smart Dustbin system.

1. **Low Cost:** The system should be affordable and use easily available components.

2. **Easy to Use:** The dustbin should operate automatically without requiring special instructions.
3. **Reliability:** The system should work properly and consistently when power is supplied.
4. **Fast Response:** The lid should open quickly when the sensor detects an object.
5. **Low Power Consumption:** The system should consume a small amount of electrical power.
6. **Easy Maintenance:** The components should be easy to check, replace, and repair.
7. **Compact Design:** The electronic components should be arranged neatly without occupying unnecessary space.
8. **Safety:** The system should be safe for users and should not have exposed electrical connections.
9. **Durability:** The components and mechanical arrangement should withstand regular use.
10. **Simple Operation:** The system should not require internet connectivity, cloud services, or advanced technology.

CHAPTER 5
CIRCUIT DESIGN AND SYSTEM WORKING

5.1 Design Requirements

The design requirements describe what the hardware must achieve so that the Smart Dustbin can detect waste, open and close its lid automatically, and report its fill level to a remote dashboard.

- Microcontroller: a compact programmable chip that reads sensors, makes decisions and controls output devices. The ESP32 is used here.
- Sensor: a device that converts a physical quantity such as distance or motion into an electrical signal.
- Actuator: a device that produces physical movement from an electrical signal. The servo motor is the actuator here.
- IoT (Internet of Things): a network of physical devices that collect and exchange data over the internet.

5.2 Components and Pin Configuration

Sr. No.	Component	Specification	Purpose
1	**ESP32 DevKit V1**	Dual core, 3.3 V logic, Wi-Fi and Bluetooth	Main controller and IoT gateway
2	**Ultrasonic Sensor HC-SR04 (Inside)**	Range 2–400 cm, 5 V	Measures waste level inside the bin
3	**IR Obstacle Sensor / Second HC-SR04 (Outside)**	Detection range about 2–30 cm	Detects a person or hand near the bin
4	**Servo Motor SG90**	5 V, torque about 1.8 kg·cm, 0°–180°	Opens and closes the lid
5	**Buzzer**	5 V active buzzer	Provides an audible alert
6	**LEDs (Green, Red)**	5 mm, with 220 Ω resistors	Provides visual status indication
7	**16×2 LCD with I2C Module**	5 V, I2C address 0x27	Displays fill percentage locally
8	**Power Supply**	5 V, 2 A adapter or power bank	Powers the complete circuit

9	**Resistors**	1 kΩ and 2 kΩ	Used as a voltage divider for the ECHO pin
10	**Jumper Wires, Breadboard/PCB**	—	Used for electrical connections

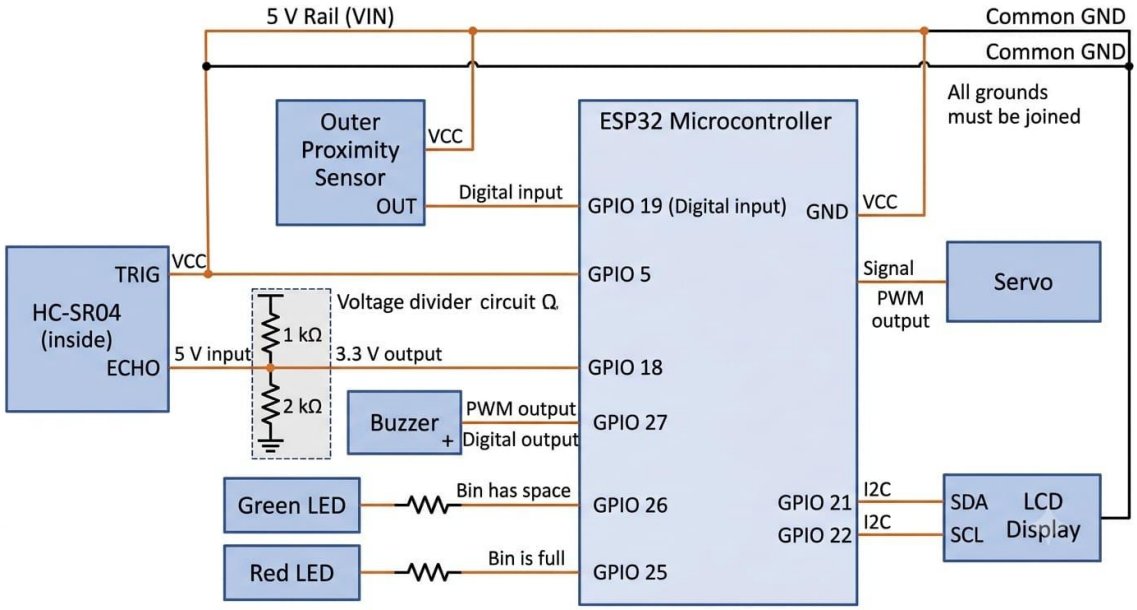


Figure 5.1: Pin Connection Diagram

5.3 Working Principle

Definition of ultrasonic distance measurement: An ultrasonic sensor sends out a 40 kHz sound pulse and measures the time taken for the echo to return after hitting an object. Distance is calculated as:

Distance (cm) = (Time in microseconds × 0.0343) / 2

The division by 2 is needed because the measured time covers the trip to the object and back.

Step by step working

1. **Power on and initialisation:** The ESP32 starts, initialises the sensors, servo and LCD, connects to Wi-Fi, and moves the lid to the closed position (0°). The green LED turns on.
2. **Detection of a user:** The outer sensor continuously checks for an object within about 30 cm. When a person or hand is detected, it sends a HIGH signal to the ESP32.
3. **Lid opening:** The ESP32 sends a PWM signal to the servo, which rotates to about 90° to 110° and lifts the lid. The lid stays open for around 5 seconds so the waste can be dropped in.
4. **Lid closing:** When no object is detected any more and the delay has passed, the servo returns to 0° and the lid closes.
5. **Waste level measurement:** After the lid closes, the inner ultrasonic sensor measures the distance from the lid to the waste surface. Measuring only with the lid

closed gives stable readings, because a moving hand or open lid would give false values.

6. **Fill percentage calculation:** With an empty bin depth of 30 cm,

Fill % = ((Bin depth − Measured distance) / Bin depth) × 100

For example, a measured distance of 6 cm gives ((30 − 6) / 30) × 100 = 80%.

7. **Local indication:**
    - Below 70%: green LED on, normal operation.
    - 70% to 90%: LCD shows a "Nearly Full" message.
    - Above 90%: red LED on, buzzer sounds, and the lid stops opening for new waste to avoid overflow, if this option is enabled.
8. **Data upload:** The ESP32 sends the fill percentage and status to the cloud dashboard over Wi-Fi every few seconds. A notification is raised for the collection staff once the bin crosses the full threshold.
9. **Reset:** After the bin is emptied, the sensor reads a large distance again, the fill value drops, and the system returns to normal mode.

5.4 Constraints

Technical constraints

- **Sensor accuracy:** HC-SR04 accuracy is about ±3 mm, but readings can fluctuate on soft, uneven or angled waste such as paper or cloth that absorbs or scatters sound. Averaging several readings reduces this error.
- **Dead zone:** The sensor cannot measure below 2 cm, so the full level threshold is set a few centimetres from the sensor.
- **Voltage level mismatch:** ESP32 GPIO pins accept only 3.3 V, so a voltage divider is mandatory on the 5 V ECHO line.
- **Power:** The servo causes current spikes at start. An underpowered supply can reset the ESP32. A stable 5 V, 2 A source is needed.
- **Wi-Fi dependency:** Remote monitoring works only where Wi-Fi coverage is available. Local indicators still work offline.
- **Environmental limits:** Moisture, dust and heat can affect the sensors. Wet waste and odour may reduce component life, so the electronics should be mounted inside a protective box on the lid.
- **Sensor range:** The outer sensor detects objects only up to about 30 cm, so users must approach closely.

CHAPTER 6
APPLICATION DESIGN

6.1 Application Modules

Definition: An application module is a self contained part of the software that performs one specific function. Dividing the system into modules makes the program easier to build, test and maintain.

6.1.1 Wi-Fi Connection and Web Server

The ESP8266 microcontroller has built-in Wi-Fi connectivity, which makes it suitable for IoT-based applications. In the Smart Dustbin project, Wi-Fi can be used to connect the system to a wireless network and enable remote monitoring.When the ESP8266 is powered on, it can be programmed to connect to a Wi-Fi network using the network name (SSID) and password. After establishing a connection, the ESP8266 can communicate with other devices over the network.Wi-Fi connectivity can be used to transmit information about the dustbin's operating status to a web interface. Additional sensors and programming can also be used to monitor the waste level and send notifications when the dustbin becomes full.

6.1.2 Distance Measurement

Distance measurement is an important part of the Smart Dustbin system. The ultrasonic sensor (HC-SR04) is used to measure the distance between the sensor and a nearby object, such as a user's hand.

The sensor works by transmitting ultrasonic waves and receiving the reflected waves from an object. The ESP8266 measures the time taken by the waves to travel to the object and return. It then calculates the distance using the speed of sound.

6.1.3 Level Calculation and Alert Logic

Level calculation is used to determine how much waste is present inside the dustbin. In an advanced Smart Dustbin system, an ultrasonic sensor can be

installed at the top of the dustbin facing downward to measure the distance between the sensor and the waste surface.

When the dustbin is empty, the measured distance is generally greater. As waste accumulates, the distance between the sensor and the waste surface decreases.

6.1.4 OLED Display

The Adafruit_SSD1306 and Adafruit_GFX libraries [9] are used to clear the display buffer, print the level percentage in large text, draw a vertical level bar and show a status line (IP address, TANK FULL!, LOW WATER! or Sensor error!). This is repeated on every update.

6.1.5 Dashboard Page and Alerts

The dashboard is a single HTML page stored in the program memory of the ESP32. Its JavaScript asks for /data and /history at the selected interval (1, 2, 5 or 10 seconds) using the fetch function [20], redraws the tank and the trend chart, and changes the banner and the colour of the tank when the level reaches the full or low threshold. When an alert starts, the page can play a beep and show a browser notification if the user has turned them on. The Export CSV button saves the recent readings, and the Calibration & alerts panel sends new settings to /settings.

6.2 User Interface Design

The user interface (UI) is the part of the system through which people view information and interact with the application. The Smart Dustbin has two interfaces.

A. Physical interface (on the dustbin)

- **Automatic lid:** Opens when a user approaches, so no touch is needed.
- **LCD display:** Shows the line "Fill Level: 65%" and a status such as "Space Available" or "Bin Full".
- **LED indicators:** Green means space is available, red means the bin is full.
- **Buzzer:** Gives a short beep sequence when the bin is full.

. **B. Dashboard interface (mobile app or web page)**

The dashboard is designed to be simple so that municipal or cleaning staff can read it at a glance. It contains:

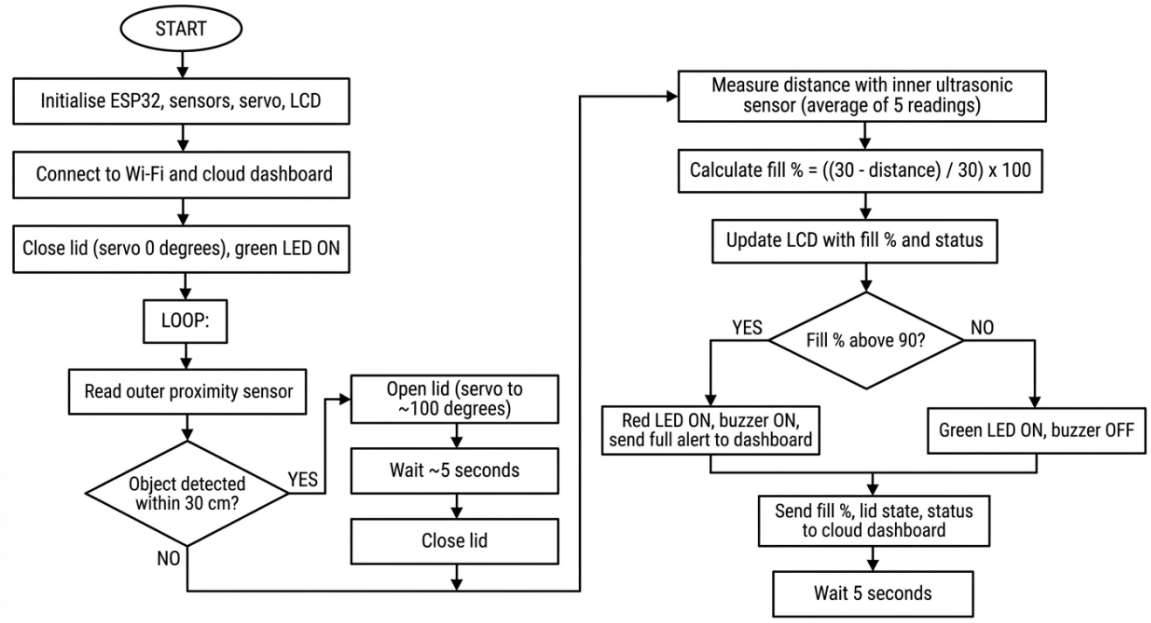
1. **Fill level gauge:** A circular or linear gauge showing the fill percentage from 0 to 100, coloured green, yellow and red.

2. **Status indicator:** A text label showing Empty, Half, Nearly Full or Full, together with an LED style widget.
3. **Line chart:** Shows the fill level over time, which helps identify how fast the bin fills.
4. **Lid status:** Shows whether the lid is currently open or closed.
5. **Alert panel and notifications:** Shows a message and sends a push notification, for example "Bin 01 is 92% full, please empty".
6. **Device status:** Shows whether the ESP32 is online and the last update time.

6.3 Dashboard Connectivity and Program Flow

The ESP32 acts as an IoT node. It connects to the local Wi-Fi router using the SSID and password stored in the program. After connecting, it communicates with a cloud IoT platform such as Blynk, ThingSpeak or Firebase. The steps are:

1. The ESP32 authenticates with the cloud using a unique token or API key.
2. Sensor data is sent through the internet over HTTP or MQTT at a fixed interval of about 5 to 10 seconds.
3. The cloud stores the values and updates the widgets bound to them. In Blynk, for example, virtual pin V0 carries the fill percentage, V1 the lid state and V2 the status text.
4. The cloud triggers a notification when the fill value crosses a set threshold.
5. If Wi-Fi is lost, the ESP32 keeps operating locally and tries to reconnect periodically.


START
Initialise ESP32, sensors, servo, LCD
Connect to Wi-Fi and cloud dashboard
Close lid (servo 0 degrees), green LED ON
LOOP:
Read outer proximity sensor
Object detected within 30 cm?
YES
NO
Open lid (servo to ~100 degrees)
Wait ~5 seconds
Close lid
Measure distance with inner ultrasonic sensor (average of 5 readings)
Calculate fill % = ((30 - distance) / 30) x 100
Update LCD with fill % and status
Fill % above 90?
YES
NO
Red LED ON, buzzer ON, send full alert to dashboard
Green LED ON, buzzer OFF
Send fill %, lid state, status to cloud dashboard
Wait 5 seconds

CHAPTER 7
RESULTS AND DISCUSSION

The Smart Dustbin successfully automates the opening and closing of the lid using an ESP8266, ultrasonic sensor, and servo motor. It reduces physical contact, improves hygiene, and makes waste disposal more convenient.

7.1 System Testing and Evaluation

System testing and evaluation is an important phase of the Smart Dustbin project. It is performed to verify whether all the hardware and software components work correctly together. The main purpose of testing is to ensure that the ultrasonic sensor detects nearby objects, the ESP8266 processes the sensor readings, the servo motor controls the lid, and the LED indicators operate according to the programmed logic.

Testing helps identify errors in the circuit connections, sensor readings, program logic, and mechanical movement of the dustbin lid. It also helps evaluate the reliability, functionality, and performance of the system under normal operating conditions.

7.2 Discussion of Results

The smart dustbin project successfully demonstrates the basic concept of automatic waste disposal. The ultrasonic sensor detects an object near the dustbin and sends a signal to control the servo motor. The servo motor opens and closes the lid automatically, reducing the need for physical contact. The LED lights provide a visual indication of the system's operation. The breadboard and jumper wires help connect all the electronic components properly. The project is simple, cost-effective, and easy to understand. It helps maintain hygiene and encourages proper waste disposal. However, the performance of the system depends on the sensor's accuracy and the proper functioning of the servo motor. Overall, the project demonstrates how basic electronic components can be used to create a useful automated system. In the future, additional features such as waste-level detection and alerts can be added to improve its functionality.

7.3 Implementation (Output Screenshots)

The smart dustbin prototype was implemented using an ultrasonic sensor, servo motor, breadboard, jumper wires, and LED lights. The ultrasonic sensor detects an object near the dustbin and triggers the servo motor to open the lid

automatically. After a short delay, the lid closes. The LED lights indicate the

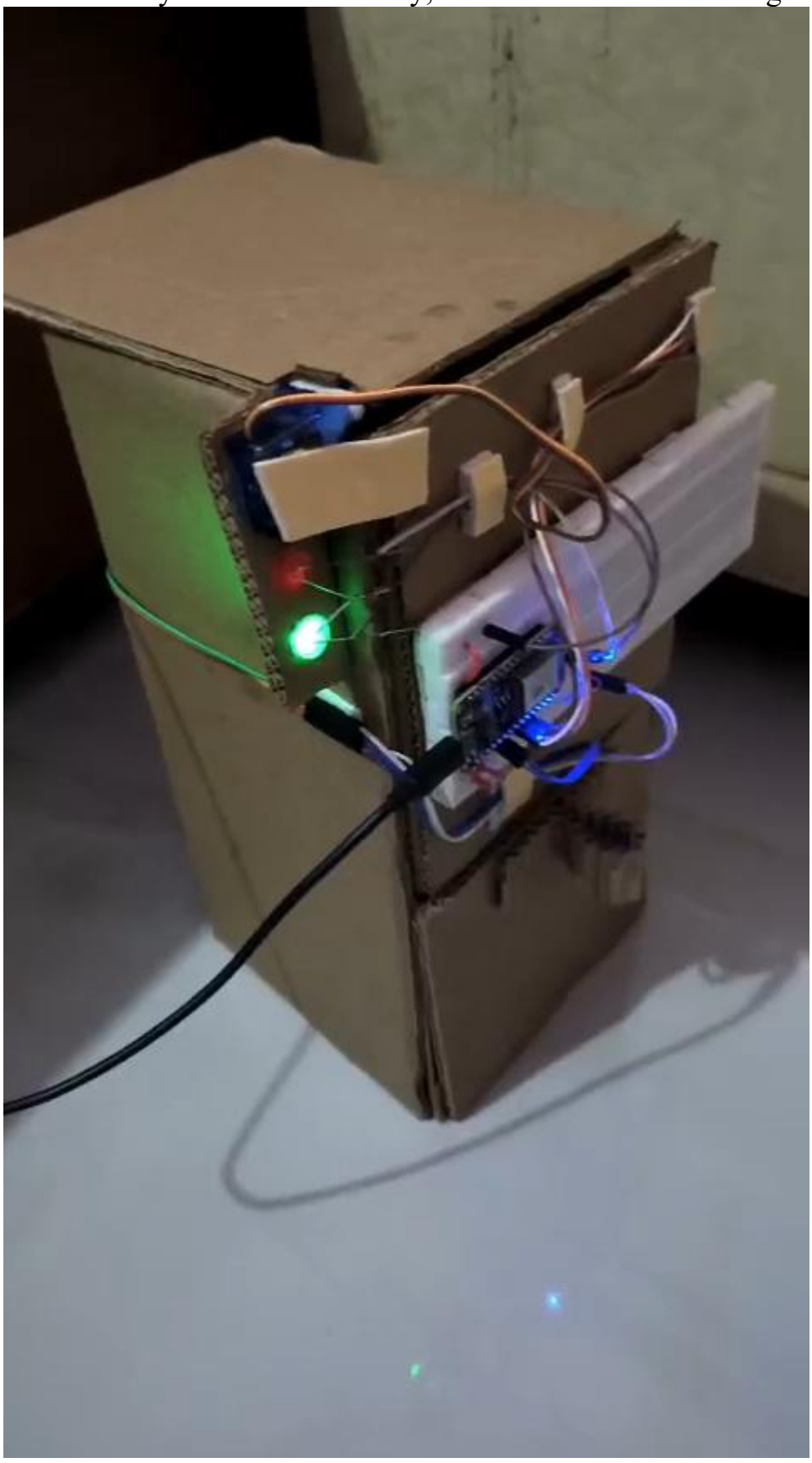


Figure 7.1: Circuit Wiring of the ES8266, Ultrasonic sensor and sarvo motor

CHAPTER 8
SUSTAINABILITY & IMPACT

8.1 Industrial Relevance

Smart dustbins are useful in industries, offices, hospitals, hotels, shopping malls, and public places. They help improve cleanliness and hygiene by reducing physical contact during waste disposal. The automatic lid makes waste disposal more convenient and encourages proper waste management. In industries, advanced smart dustbins can include waste-level sensors and alerts to improve waste collection efficiency. Our basic model demonstrates the use of automation in waste management and can be further developed for industrial applications.

8.2 Societal/Environmental Impact

The smart dustbin has a positive impact on society and the environment by promoting cleanliness, hygiene, and proper waste disposal. Its automatic lid reduces the need to touch the dustbin, helping minimize direct contact with germs and dirty surfaces. This makes waste disposal more convenient in homes, schools, colleges, hospitals, offices, and public places.From an environmental perspective, the smart dustbin encourages people to dispose of waste responsibly and helps maintain clean surroundings. Proper waste disposal can reduce littering and contribute to a healthier environment. The project also creates awareness about the importance of cleanliness and sustainable waste management. Our basic smart dustbin uses an ultrasonic sensor, servo motor, breadboard, jumper wires, and LED lights to demonstrate automatic operation. These components help students understand how technology can solve everyday problems. In the future, the system can be improved by adding waste-level detection, alerts, and waste segregation features. These improvements could make waste collection more efficient and reduce environmental pollution. Overall, the smart dustbin supports a cleaner, healthier, and more environmentally responsible society

8.3 SDG Mapping

The project contributes to the United Nations Sustainable Development Goals (SDGs) [10] as shown in Table 8.1.

Table 8.1: Mapping of smartdustbin to the Sustainable Development Goals

SDG	Contribution of smartdustbin
• SDG 3	Good Health and Well-being: Reduces direct contact with the dustbin and promotes hygiene.
• SDG 11	Sustainable Cities and Communities: Encourages cleaner surroundings and better waste disposal in public places.
• SDG 12	Responsible Consumption and Production: Promotes responsible waste disposal and awareness of waste management.
• SDG 13	Climate Action: Supports environmental awareness and sustainable waste management practices.

CHAPTER 9
CONCLUSION AND FUTURE SCOPE

9.1 Summary of Findings

The smart dustbin project demonstrates the basic concept of automatic waste disposal using an ultrasonic sensor and servo motor. The sensor detects an object near the dustbin, and the servo motor automatically opens and closes the lid. LED lights indicate the system's operation.

The project shows that simple and affordable electronic components can make waste disposal more convenient and hygienic. It reduces the need for physical contact and encourages proper waste disposal. The prototype also provides practical knowledge of electronics and automation.

Overall, the project is a useful step toward cleaner surroundings and smarter waste management. In the future, features such as waste-level detection and automatic alerts can be added to improve its performance.

9.2 Possible Enhancements

1. Waste-Level Detection: Add a sensor to detect when the dustbin is full.
2. Mobile Alerts: Send notifications when the bin needs to be emptied.
3. IoT Integration: Connect the dustbin to the internet for remote monitoring.
4. Solar Power: Use solar energy to reduce electricity consumption.
5. Waste Segregation: Separate wet and dry waste automatically.
6. Improved Sensors: Use more accurate sensors for better object detection.
7. Battery Backup: Add a rechargeable battery for uninterrupted operation.
8. Improved Design: Make the dustbin stronger, more attractive, and durable.
9. Odour Control: Add a system to reduce unpleasant smells.
10. Larger Capacity: Design bigger dustbins for public places and industries.

REFERENCES

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APPENDIX

Appendix A: Development Approach

The smart dustbin was developed by selecting basic electronic components, including an ultrasonic sensor, servo motor, breadboard, jumper wires, and LED lights. The components were connected properly to create the circuit. The sensor detects objects near the dustbin, and the servo motor controls the lid automatically. The system was tested to check its proper functioning. In the future, additional features can be included to improve its performance and efficiency.

Appendix B: Technology Stack Summary

The smart dustbin uses an ultrasonic sensor to detect nearby objects and a servo motor to open and close the lid automatically. A breadboard and jumper wires connect the components, while LED lights indicate operation. The system uses basic electronics and programming to demonstrate automatic waste disposal.

Appendix C: Firmware (Arduino Sketch)

The firmware controls the working of the smart dustbin using Arduino programming. The code receives signals from the ultrasonic sensor to detect nearby objects and operates the servo motor to open and close the lid automatically. LED lights indicate the system's operation. The program helps the components work together to provide automatic and convenient waste disposal.

```
#include <Servo.h>


#define TRIG_PIN   D6

#define ECHO_PIN   D7


#define SERVO_PIN  D4


#define RED_LED    D2

#define GREEN_LED  D1


Servo dustbinServo;


// SG90 positions

const int CLOSED_ANGLE = 10;

const int OPEN_ANGLE   = 130;
```

```
// Ultrasonic detection
const int DETECT_DISTANCE = 4;
const int RESET_DISTANCE  = 30;

// Open for 5 seconds
const unsigned long OPEN_TIME = 5000;

// Red blink speed
const unsigned long BLINK_TIME = 100;

bool binOpen = false;
bool readyForNewHand = true;

unsigned long openStartTime = 0;
unsigned long lastBlinkTime = 0;

bool redState = false;



// ------------------------------------------------
// GET DISTANCE
// ------------------------------------------------

long getDistance() {

  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(3);

  digitalWrite(TRIG_PIN, HIGH);
  delayMicroseconds(10);

  digitalWrite(TRIG_PIN, LOW);

  long duration = pulseIn(ECHO_PIN, HIGH, 30000);

  if (duration == 0) {
```

```
    return 999;
  }

  long distance = duration * 0.0343 / 2;

  return distance;
}



// ------------------------------------------------
// OPEN BIN
// ------------------------------------------------

void openBin() {

  Serial.println("Opening bin...");

  dustbinServo.write(OPEN_ANGLE);

  binOpen = true;
  readyForNewHand = false;

  openStartTime = millis();

  digitalWrite(GREEN_LED, LOW);

  redState = true;
  digitalWrite(RED_LED, HIGH);

  lastBlinkTime = millis();
}



// ------------------------------------------------
// CLOSE BIN
// ------------------------------------------------
```

```
void closeBin() {

  Serial.println("Closing bin...");

  dustbinServo.write(CLOSED_ANGLE);

  binOpen = false;

  digitalWrite(RED_LED, LOW);
  digitalWrite(GREEN_LED, HIGH);
}



// ------------------------------------------------
// SETUP
// ------------------------------------------------

void setup() {

  Serial.begin(115200);

  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);

  pinMode(RED_LED, OUTPUT);
  pinMode(GREEN_LED, OUTPUT);

  digitalWrite(TRIG_PIN, LOW);

  // SG90 attach with safe pulse range
  dustbinServo.attach(SERVO_PIN, 500, 2400);

  // Start position
  dustbinServo.write(CLOSED_ANGLE);

  digitalWrite(RED_LED, LOW);
  digitalWrite(GREEN_LED, HIGH);
```

```
  delay(500);

  Serial.println("================================");
  Serial.println(" SMART DUSTBIN READY");
  Serial.println("================================");
}



// ------------------------------------------------
// LOOP
// ------------------------------------------------

void loop() {

  long distance = getDistance();

  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");



  // ================================================
  // BIN CLOSED
  // ================================================

  if (!binOpen) {

    digitalWrite(GREEN_LED, HIGH);

    // Hand detected
    if (readyForNewHand && distance <= DETECT_DISTANCE) {

      Serial.println("Hand detected!");

      openBin();
    }
```

```
    // Hand moved away
    if (!readyForNewHand && distance >= RESET_DISTANCE) {

      readyForNewHand = true;

      Serial.println("Ready for next hand.");
    }
  }


  // ================================================
  // BIN OPEN
  // ================================================

  if (binOpen) {

    // Fast red blink
    if (millis() - lastBlinkTime >= BLINK_TIME) {

      lastBlinkTime = millis();

      redState = !redState;

      digitalWrite(RED_LED, redState);
    }


    // 5 seconds complete
    if (millis() - openStartTime >= OPEN_TIME) {

      closeBin();
    }
  }

  delay(50);
}
```

Appendix D: Wiring Summary

Table D.1: ESP32 Wiring Summary

ESP8266 Pin	Connected To
VN/5V	HC-SR04 VCC (5v supply)
D5(GPIO 14)	HC-SR04 TRIG
D6(GPIO 12)	HC-SR04 ECHO (through a voltage divider)
D7(GPIO 13)	SERVO MOTOR SINGLE WIRE
3.3V	OLED VCC
D2(GPIO 4)	OLED SDA
GND	HC-SR04 GND, OLED GND AND SARVO MOTOR (common ground)